



SHENTON
COLLEGE

Year 11 Mathematics Applications

Investigation 1 2019

Calculator Assumed

Name: **SOLUTIONS**

Teacher: Cheshire Friday Hill Le Loh Staffe

Reading Time: 5 minutes
Working Time: 45 minutes

Marks	/38
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Materials allowed: Formula sheet, Classpad, calculator.

All necessary working and reasoning must be shown for full marks.

-1 units
-1 rounding

Reducible Interest Loans

Congratulations! You have just passed your practical driving assessment and now get to buy your very first car. You have worked part-time for several years and have saved enough money for a deposit but need to get a small car loan to afford the first car of your dreams.

In this investigation, you will consider **three different scenarios for the same loan** to investigate how making different repayment amounts effects the outcome of a reducible interest loan.

Loan Details:

You have taken out a car loan of \$10 000 and have agreed to pay a fixed interest rate of 7.5% p.a. compounded monthly for 1 year.

Scenario 1: You make the standard monthly repayments of \$867.58 until the loan is paid off.

Scenario 2: You land a new full-time job *before* you take out the loan. This allows you to negotiate the terms of the loan so that you are able to make **higher** monthly repayments until the loan is paid off.

Scenario 3: You move out of home *before* you take out the loan. You negotiate the terms of the loan so that you are able to make **lower** repayments until the loan is paid off.

Question 1. [3 marks]

If you were to make no repayments, calculate the total amount of interest that would be charged at the end of the loan period.

$$A = 10\,000 \left(1 + \frac{0.075}{12}\right)^{12} \quad \checkmark \text{ substitute}$$

$$= \$10\,776.33 \quad \checkmark \text{ final amount}$$

$$I = \$776.33 \quad (2d.p.) \quad \checkmark \text{ subtract principal}$$

SCENARIO 1:**Question 2. [8 marks – 2, 2, 2, 2]**

In reality, the monthly repayments that you make will reduce the principal and therefore the total amount of interest charged throughout the loan period. This type of loan is called a reducible interest loan.

In reducible interest loans, the interest is added to the opening balance each month before the repayment is taken off to calculate the closing balance.

i.e. **Closing Balance = Opening balance + Interest – Repayment**

- a) Given that the annual interest rate is 7.5%, calculate the monthly interest rate. Do not round your answer.

$$\frac{7.5}{12} = 0.625\% \text{ or } 0.00625 \quad \checkmark \text{ divide } r \text{ by } 12 \quad \checkmark \text{ correct monthly interest rate}$$

- b) Given that your opening balance is \$10 000, calculate the amount of interest that is accrued after 1 month.

$$10\,000 \times 0.00625 = \$62.50 \quad \checkmark \text{ multiply principal by } 0.00625 \quad \checkmark \text{ correct interest}$$

- c) What is the total amount owing after the first month, just before the first repayment is made?

$$\$10\,000 + 62.50 = \$10\,062.50 \quad \checkmark \text{ add interest to principal} \quad \checkmark \text{ correct total}$$

- d) If you make a monthly repayment of \$867.58, what is the closing balance i.e. how much will you owe immediately after making the first repayment?

$$\$10\,062.50 - 867.58 = \$9\,194.92 \quad \checkmark \text{ subtract repayment} \quad \checkmark \text{ correct closing balance}$$

Question 3. [6 marks – 1, 4, 1]

At the beginning of a loan period a large proportion of the repayment that you make goes towards paying the interest whilst the remainder goes towards paying the principal. Gradually the proportion of the repayment that goes towards paying the interest reduces whilst the proportion of the repayment that goes towards paying the principal increases. This process is called *amortisation* and continues until both the principal and interest are paid off completely.

We will now model the amortisation process in the table below.

✓ row 1 complete

- Complete the cells in the first row of the table using the results you obtained in Question 2.
- Confirm that the loan will take 12 months to pay by filling out the last 3 rows of the table.

Month	Opening Balance (\$)	Interest accrued this month (\$)	Monthly Repayment (\$)	Closing Balance (\$)
1	\$10 000	\$62.50	\$867.58	\$9194.92
2	\$9,194.92	\$57.47	\$867.58	\$8,384.81
3	\$8,384.81	\$52.41	\$867.58	\$7,569.63
4	\$7,569.63	\$47.31	\$867.58	\$6,749.36
5	\$6,749.36	\$42.18	\$867.58	\$5,923.97
6	\$5,923.97	\$37.02	\$867.58	\$5,093.41
7	\$5,093.41	\$31.83	\$867.58	\$4,257.67
8	\$4,257.67	\$26.61	\$867.58	\$3,416.70
9	\$3,416.70	\$21.35	\$867.58	\$2,570.47
10	\$2570.47	\$16.07	\$867.58	\$1718.96
11	\$1718.96	\$10.74	\$867.58	\$862.12
12	\$862.12	\$5.39	\$867.58	-\$0.07

✓ opening balance ✓ interest (867.58) ✓ repayment (0) ✓ closing balance

- What is the total amount of interest paid throughout the loan period?

\$410.89

\$410.88

✓ sum of interest column

OR (11 Months x 867.58 + final repay - 10000)

6

SCENARIO 2:

Question 4. [8 marks – 1, 5, 1, 1]

You land a full-time job and find that you can make larger repayments towards your car loan.

- a) If you can actually afford to pay \$275 per week, calculate your new monthly repayment to two decimal places.

$$\frac{275 \times 52}{12} = \$1191.67 \text{ (2 d.p.)}$$

✓ correctly convert weekly to monthly (2 d.p.)

- b) Complete the table below to model the repayment schedule given that you can afford to pay \$275 per week towards your car loan.

Month	Opening Balance (\$)	Interest accrued this month (\$)	Monthly Repayment (\$)	Closing Balance (\$)
1	\$10 000	\$62.50	\$1191.67	\$8870.83
2	\$8,870.83	\$55.44	\$1191.67	\$7,734.60
3	\$7,734.60	\$48.34	\$1191.67	\$6,591.27
4	\$6,591.27	\$41.20	\$1191.67	\$5,440.80
5	\$5,440.80	\$34.00	\$1191.67	\$4,283.13
6	\$4,283.13	\$26.77	\$1191.67	\$3,118.23
7	\$3118.23	\$19.49	\$1191.67	\$1946.05
8	\$1946.05	\$12.16	\$1191.67	\$766.55
9	\$766.55	\$4.79	\$1191.67	-\$420.33

- c) What is the total amount of interest paid throughout the loan period?
- ✓ month ✓ opening balance ✓ interest ✓ repayment ✓ closing balance

$$\$304.70 \quad \checkmark \quad \text{sum of interest column}$$

\$304.69

- d) How will your final repayment be affected by the final closing balance?

The final repayment will be \$420.33 less than \$1191.67 i.e. \$771.34.

✓ recognise reduced payment

SCENARIO 3:

Question 5. [6 marks – 4, 1, 1]

You move out of home and need to budget a major portion of your monthly pay towards rent, bills and living costs. You negotiate the terms of your loan with your bank so that you are able to make smaller repayments.

- a) You have budgeted \$710 per month to make repayments towards your car loan. The repayment schedule for the first year of your loan is displayed below. Complete the last 3 rows of the table.

Month	Opening Balance (\$)	Interest accrued this month (\$)	Monthly Repayment (\$)	Closing Balance (\$)
1	\$10,000.00	\$62.50	\$710.00	\$9,352.50
2	\$9,352.50	\$58.45	\$710.00	\$8,700.95
3	\$8,700.95	\$54.38	\$710.00	\$8,045.33
4	\$8,045.33	\$50.28	\$710.00	\$7,385.62
5	\$7,385.62	\$46.16	\$710.00	\$6,721.78
6	\$6,721.78	\$42.01	\$710.00	\$6,053.79
7	\$6,053.79	\$37.84	\$710.00	\$5,381.62
8	\$5,381.62	\$33.64	\$710.00	\$4,705.26
9	\$4,705.26	\$29.41	\$710.00	\$4,024.67
10	\$4,024.67	\$25.15	\$710.00	\$3,339.82
11	\$3,339.82	\$20.87	\$710.00	\$2,650.70
12	\$2,650.70	\$16.57	\$710.00	\$1,957.26
13	\$1,957.26	\$12.23	\$710.00	\$1,259.50 ^{.49}
14	\$1,259.50 ^{.49}	\$7.87	\$710.00	\$557.37 ^{.36}
15	\$557.37 ^{.36}	\$3.48	\$710.00	-\$149.15

✓ opening
balance

✓ interest

✓ repayment

✓ closing
balance

(4)

b) What is the total amount of interest paid throughout the loan period?

$\$500.85$ ✓ sum of interest column
 $.84$

e) How will your final repayment be affected by the final closing balance?

The final repayment will be $\$149.15$ less than $\$710$ i.e. $\$560.85$.

✓ recognise reduced repayment

Question 6. [3 marks – 2, 1]

Complete the following table to help you compare Scenario 1, Scenario 2 and Scenario 3.

Scenario 1	Scenario 2	Scenario 3
Monthly repayment: $\$867.58$	Monthly repayment: $\$1191.67$	Monthly repayment: $\$710$
Total interest or repayment: ↙ ↘ $\$410.89$ $\$10410.89$	Total interest or repayment: ↙ ↘ $\$304.70$ $\$10304.70$	Total interest or repayment: ↙ ↘ $\$500.85$ $\$10500.85$
Length of loan: 12 months	Length of loan: 9 months	Length of loan: 15 months

a) What conclusions can you make about the effects of different repayment amounts on the length of the loan and the interest accrued?

- The higher the repayment, the less interest accrued and the shorter the period of the loan ✓ effect of ↑ repayment on length of loan and interest
- The lower the repayment, the greater the interest accrued and the longer the length of the loan. ✓ effect of ↓ repayment on length of loan and interest

b) Which scenario is financially better for you and why?

Scenario 2 because you pay less interest in total.

✓ choose correct scenario with reason

Question 7. [4 marks – 1, 1, 1, 1]

State how changing the following could affect a person's financial situation:

- a) Increasing the interest rate of a car loan.

Increasing the interest rate of a car loan will increase the amount of interest accrued or increase the length of the loan if the repayment stays the same. ✓ recognise ↑ interest or length of loan

- b) Increasing the interest rate of a savings investment account.

Increasing the interest rate of a savings account will increase the amount of interest earned. ✓ recognise ↑ interest earned

- c) Increasing the amount of monthly repayment on a loan.

Increasing the amount of monthly repayment on a loan will decrease the amount of interest accrued and length of loan given that all other factors stay the same. ✓ recognise ↓ length of loan or interest accrued

- d) Increasing the length of any loan.

Increasing the length of a loan will increase the interest accrued. ✓ recognise ↑ interest

